Local Research Asset Extraction

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Abstract

We present a local document analysis system that preserves scientific figures and tables. The method detects complete vector diagrams and associates captions with structured table data. All experiments use synthetic documents created for reproducible software testing.

1 Introduction

Research papers contain visual evidence that is difficult to find with text search alone. Our approach retains source coordinates and extracts reusable figures and tables. Figure 1 illustrates the analysis pipeline, while Table 1 reports example results.

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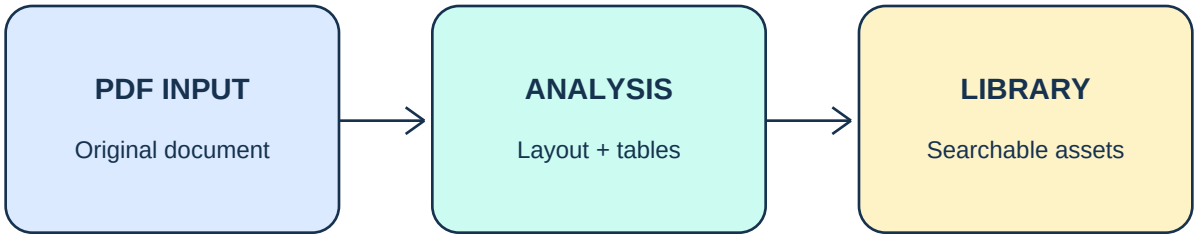

Figure 1: A fully local pipeline for extracting reusable research assets.

2 Results

The proposed workflow stores structured values as well as a faithful image of each table.

Table 1: Invented values for testing table extraction only.

Method	Figures	Tables	Example value
Method A	8	6	0.70
Method B	10	9	0.95
Reference	10	10	1.00

3 Discussion

Every extracted asset remains connected to its source page. Users may correct a caption or crop without changing the source PDF. This benchmark is intentionally small and does not represent the extraction quality of arbitrary academic papers.

References

[1] Chen, A. and Smith, B. Local Research Asset Extraction. Synthetic Test Proceedings, 2026.